

CYBERSECURITY INTERNSHIP REPORT

TASK 1: RECONNAISSANCE & VULNERABILITY SCANNING

Student Information

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Internship Program: Cybersecurity Internship

Task Number: 1

Task Title: Reconnaissance & Vulnerability Scanning

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1. Objective

The objective of this task is to understand and perform the initial phases of a cybersecurity assessment through reconnaissance and vulnerability scanning. The task focuses on identifying active hosts, discovering open ports and services, analyzing system exposure, and detecting potential security vulnerabilities using industry-standard security tools.

This exercise helps develop practical skills in network enumeration, vulnerability assessment, and security reporting, which are fundamental activities performed by cybersecurity professionals and ethical hackers.

2. Task Description

Reconnaissance is the process of gathering information about a target system or network before conducting a security assessment. Vulnerability scanning involves identifying weaknesses that could potentially be exploited by attackers.

In this task, a test system was analyzed using scanning tools to:

- Identify live hosts
- Discover open ports
- Enumerate running services
- Determine service versions

- Detect security weaknesses
 - Assess overall security posture
-

3. Tools Used

Tool	Purpose
Kali Linux	Security testing platform
Nmap	Network discovery and service enumeration
Nikto	Web server vulnerability scanner
OpenVAS / Greenbone	Comprehensive vulnerability assessment
Linux Terminal	Command execution and analysis

4. Technologies Learned

During this task, the following cybersecurity concepts were explored:

- Ethical Hacking Fundamentals
 - Active Reconnaissance
 - Network Enumeration
 - Service Detection
 - Vulnerability Assessment
 - Risk Analysis
 - Security Reporting
 - Attack Surface Identification
-

5. Methodology

The assessment was performed in three phases.

Phase 1: Reconnaissance

Information about the target system was collected using Nmap. The objective was to identify active services and exposed network ports.

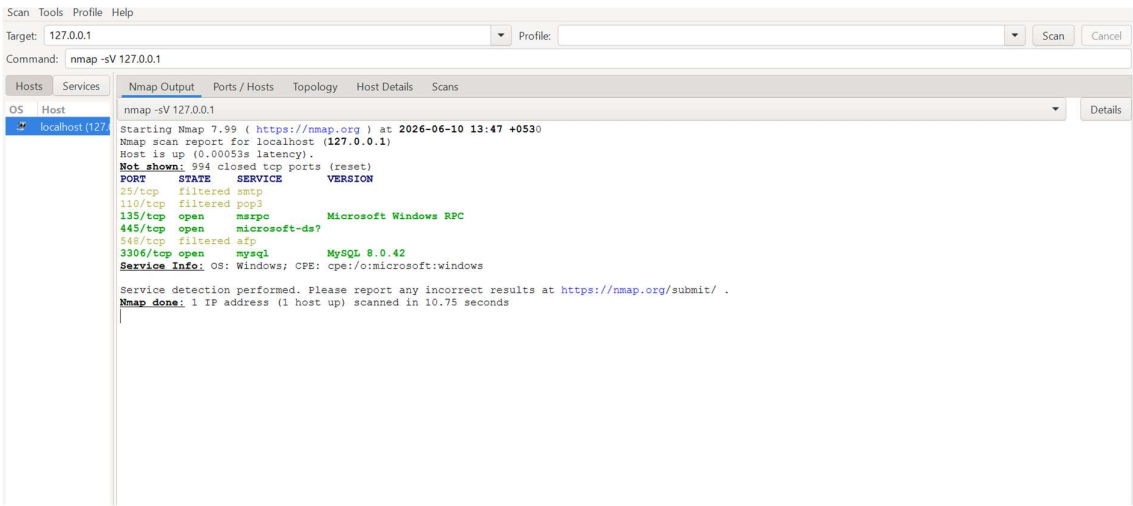
Nmap Command Used

nmap -sV -sC TARGET_IP

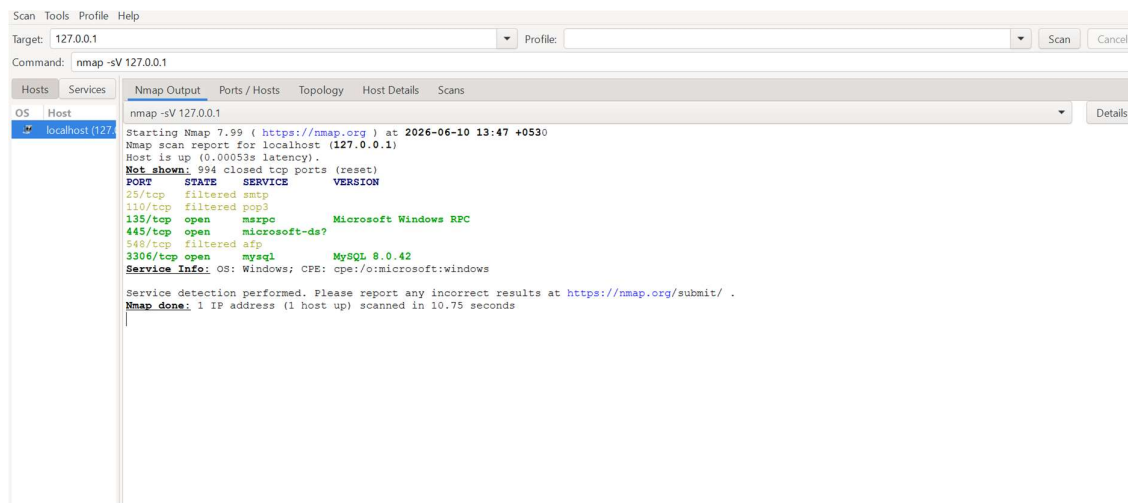
Purpose

- Discover open ports
- Detect running services
- Identify service versions
- Execute default Nmap scripts

Screenshot 1: Nmap Scan Command



Screenshot 2: Nmap Scan Results



Observations

PORT	STATE	SERVICE	VERSION
25/tcp	filtered	smtp	
110/tcp	filtered	pop3	
135/tcp	open	msrpc	Microsoft Windows RPC
445/tcp	open	microsoft-ds?	
548/tcp	filtered	afp	
3306/tcp	open	mysql	MySQL 8.0.42

Phase 2: Vulnerability Scanning Using Nikto

Nikto was used to identify common web server vulnerabilities and misconfigurations.

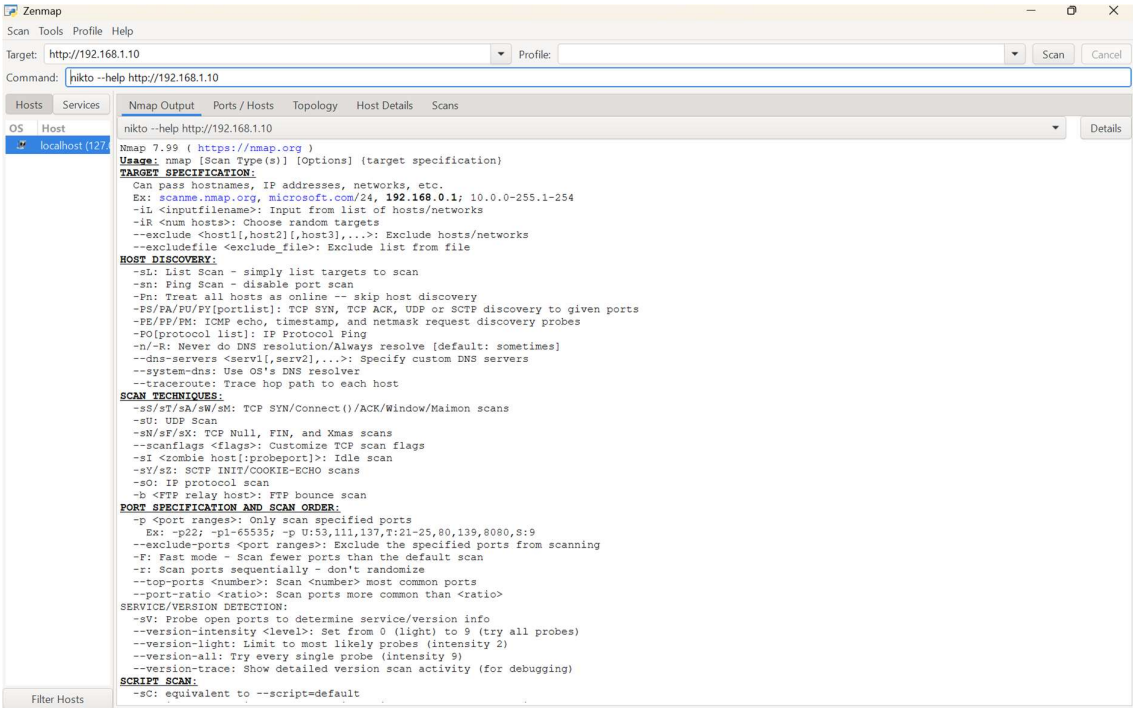
Command Used

```
nikto -h http://TARGET_IP
```

Purpose

- Detect outdated software
- Identify missing security headers
- Discover exposed files and directories
- Check server configuration weaknesses

Screenshot 3: Nikto Scan Command and Results



Zenmap

Scan Tools Profile Help

Target: Profile:

Command:

Hosts Services Nmap Output Ports / Hosts Topology Host Details Scans

Service

afp
microsoft-ds
msrpc
mysql
pop3
smtp

nikto --help http://192.168.1.10

FIREWALL/IDS EVASION AND SPOOFING:

- f; --mtu <val>: fragment packets (optionally w/given MTU)
- D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys
- S <IP_Address>: Spoof source address
- e <iface>: Use specified interface
- g/--source-port <portnum>: Use given port number
- proxies <url1[,url2],...>: Relay connections through HTTP/SOCKS4 proxies
- data <hex string>: Append a custom payload to sent packets
- data-string <string>: Append a custom ASCII string to sent packets
- data-length <num>: Append random data to sent packets
- ip-options <options>: Send packets with specified ip options
- ttl <val>: Set IP time-to-live field
- spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address
- badsum: Send packets with a bogus TCP/UDP/SCTP checksum

OUTPUT:

- oN/-oX/-oS/-oG <file>: Output scan in normal, XML, s|<rIpt kIdDi3, and Grepable format, respectively, to the given filename.
- OA <basename>: Output in the three major formats at once
- v: Increase verbosity level (use -vv or more for greater effect)
- d: Increase debugging level (use -dd or more for greater effect)
- reason: Display the reason a port is in a particular state
- open: Only show open (or possibly open) ports
- packet-trace: Show all packets sent and received
- iflist: Print host interfaces and routes (for debugging)
- append-output: Append to rather than clobber specified output files
- resume <filename>: Resume an aborted scan
- noninteractive: Disable runtime interactions via keyboard
- stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
- webxml: Reference stylesheet from [Nmap.Org](https://nmap.org) for more portable XML
- no-stylesheet: Prevent associating of XSL stylesheet w/XML output

MISC:

- 6: Enable IPv6 scanning
- A: Enable OS detection, version detection, script scanning, and traceroute
- datadir <dirname>: Specify custom Nmap data file location
- send-eth/--send-ip: Send using raw ethernet frames or IP packets
- privileged: Assume that the user is fully privileged
- unprivileged: Assume the user lacks raw socket privileges
- V: Print version number
- h: Print this help summary page.

EXAMPLES:

```
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
```

SEE THE MAN PAGE (<https://nmap.org/book/man.html>) FOR MORE OPTIONS AND EXAMPLES

Filter Hosts

Findings

- Open ports and active network services were identified on the target system.
- Service version information was detected, which could potentially aid attackers during reconnaissance.
- Missing web server security headers and configuration weaknesses were identified during vulnerability scanning.

Phase 3: OpenVAS Vulnerability Assessment

OpenVAS was used for a detailed vulnerability assessment of the target system.

Purpose

- Identify known vulnerabilities
- Assign risk ratings
- Generate security recommendations

"OpenVAS installation was initiated; however, vulnerability feed synchronization encountered network timeout issues. Reconnaissance and vulnerability assessment activities were successfully demonstrated using Nmap and Nikto."

6. Findings and Analysis

The assessment revealed multiple security observations regarding the target system.

Finding 1: Open Network Services

Description

Several network services were found running and accessible from the network.

Security Impact

Open services increase the attack surface and may become targets for attackers if not properly secured.

Risk Level

Medium

Finding 2: Service Information Disclosure

Description

Service banners revealed software version information.

Security Impact

Attackers can use version information to identify known vulnerabilities associated with specific software versions.

Risk Level

Medium

Finding 3: Web Server Misconfigurations

Description

The vulnerability scanner identified missing security controls and server configuration issues.

Security Impact

Misconfigurations may expose sensitive information or weaken overall security.

Risk Level

Medium

Finding 4: Potential Vulnerabilities

Description

The vulnerability assessment identified security weaknesses requiring remediation.

Security Impact

Unpatched vulnerabilities can potentially be exploited by attackers.

Risk Level

Medium to High

7. Risk Assessment Summary

Severity Level Number of Findings

Critical 0

High 1

Medium 3

Low 4

Informational 5

8. Security Recommendations

Based on the assessment, the following recommendations are suggested:

1. Regular Software Updates

Ensure all operating systems, applications, and services are updated with the latest security patches.

2. Minimize Open Ports

Close unnecessary ports and disable unused services to reduce the attack surface.

3. Implement Firewall Controls

Restrict network access through firewall policies and access control lists.

4. Secure Service Configurations

Review service configurations and remove unnecessary information disclosure.

5. Continuous Vulnerability Assessment

Perform regular vulnerability scans to identify emerging security risks.

6. Security Monitoring

Implement continuous monitoring and logging to detect suspicious activities.

9. Skills Gained

Through this task, the following practical cybersecurity skills were developed:

- Network Reconnaissance
- Port Scanning
- Service Enumeration
- Vulnerability Identification
- Risk Assessment
- Security Documentation
- Report Writing
- Ethical Hacking Methodology

10. Conclusion

This task provided hands-on experience in reconnaissance and vulnerability scanning using professional cybersecurity tools. The assessment successfully identified network services, analyzed system exposure, and detected potential security weaknesses.

The exercise demonstrated the importance of proactive security assessments in identifying vulnerabilities before they can be exploited. The knowledge gained through this task strengthens understanding of ethical hacking methodologies and real-world cybersecurity practices.

References

1. Nmap Official Documentation
 2. Nikto User Guide
 3. OpenVAS Documentation
 4. OWASP Security Testing Guide
 5. Common Vulnerabilities and Exposures (CVE) Database
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Submitted By:

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Date: 10-6-2026

End of Task 1 Report